

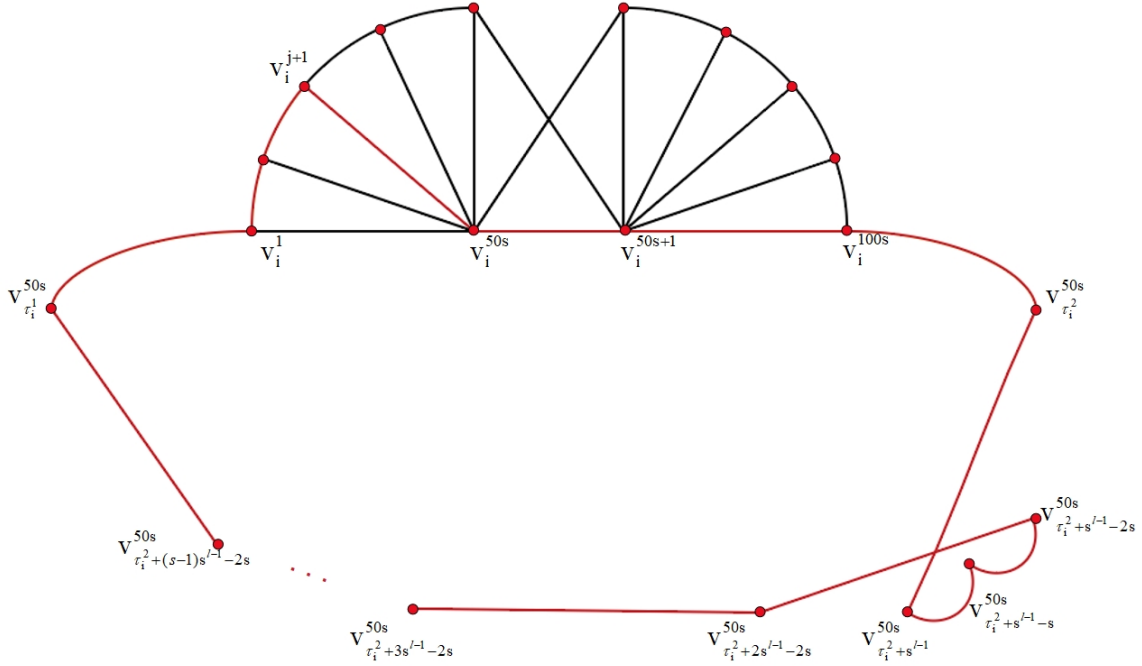


Figure 8: When $1 \leq j \leq 25s$, the edge $v_i^j v_i^{j+1}$ of G lies in a cycle of each length in $[3, 600s]$

Consider the edges $v_i^1 v_j^1, v_i^{50s} v_j^1 \in E_4$, where $1 \leq i, j \leq s^\ell$. For each $p \in [3, 50s]$,

$$v_j^1 v_i^1 v_i^2 \dots v_i^{p-2} v_i^{50s} v_j^1$$

is a cycle of length p containing $v_i^1 v_j^1$ and $v_i^{50s} v_j^1$ (see Figure 7).

Consider the edges $v_i^{50s} v_j^{50s}, v_i^{50s+1} v_j^{50s} \in E_4$, where $1 \leq i, j \leq s^\ell$. For each $4 \leq p \leq 50s$,

$$v_j^{50s} v_i^{50s} v_i^{50s+2} \dots v_i^{50s+p-2} v_i^{50s+1} v_j^{50s}$$

is a cycle of length p containing $v_i^{50s} v_j^{50s}$ and $v_i^{50s+1} v_j^{50s}$ (see Figure 7). And $v_j^{50s} v_i^{50s} v_i^{50s+1} v_j^{50s}$ is a C_3 containing both edges (see Figure 7).

Then we can use the similar cycle C^* as in Claim 8 and the similar replacement to prove every edge in E_4 is contained in a cycle of length k if $j \notin \{\tau_i^2, \tau_i^2 + s^{\ell-1}, \tau_i^2 + s^{\ell-1} - s, \tau_i^2 + s^{\ell-1} - 2s, \tau_i^2 + 2s^{\ell-1} - 2s, \dots, \tau_i^1\}$, where $3 \leq k \leq 600s$. When $j \in \{\tau_i^2, \tau_i^2 + s^{\ell-1}, \tau_i^2 + s^{\ell-1} - s, \tau_i^2 + s^{\ell-1} - 2s, \tau_i^2 + 2s^{\ell-1} - 2s, \dots, v_{\tau_i^1}\}$, we can replace τ_i^2 with $\hat{\tau}_i^2 := \tau_i^2 + s$ and add a vertex $v_{\hat{\tau}_i^2+s^{\ell-1}-3s}$ between $v_{\hat{\tau}_i^2+s^{\ell-1}-2s}$ and $v_{\hat{\tau}_i^2+2s^{\ell-1}-2s}$ in C^* . Then we use a similar replacement to prove such edges in E_4 are contained in a cycle of length k , $3 \leq k \leq 600s$. ■

Claim 10. Every edge in E_3 lies in a cycle of length k , $3s^\ell + 100s \leq k \leq v(G)$.

Proof of Claim 10. Notice that for every $3 \leq p \leq 100s - 1$, there is a path of length p contained in the copy of $H(s)$ in G connecting v_i and v_{i+1} , where $i \in [s^\ell]$ and $v_i, v_{i+1} \in V(G_1)$. Moreover, every edge in E_3 lies in a path of length $100s - 1$ contained in the copy of $H(s)$ in G connecting v_j and v_{j+1} for some $j \in [s^\ell]$, where $v_j, v_{j+1} \in V(G_1)$. Thus using the Hamilton cycle in G_1 , we can find a cycle of each length k , $k \geq 3s^\ell + 100s$ containing e for every $e \in E_3$. ■